



Error Correction Coding for TDRSS Demand Access System

ENGINEERING

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Abstract

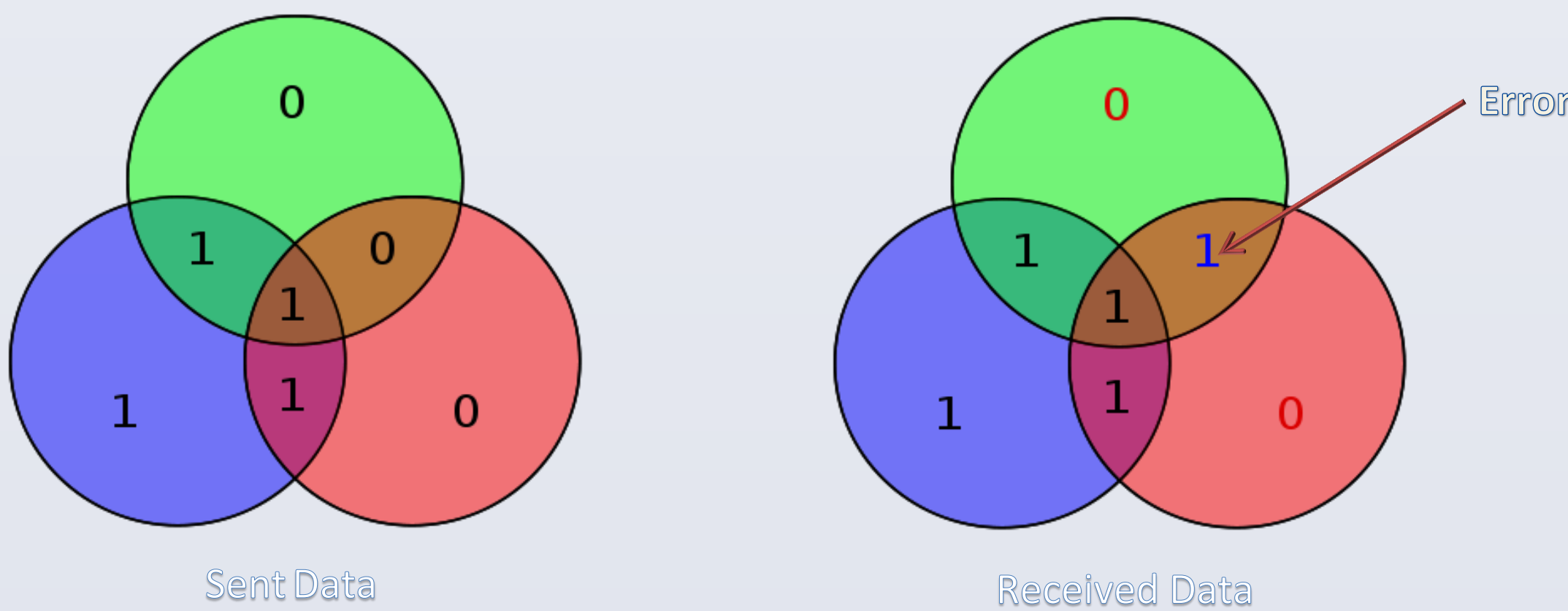
Tracking and Data Relay Satellite System (TDRSS) is a fleet of nine satellites that provide telecommunication services to NASA and several costumers. In order to provide high reliability and high data rates TDRSS is capable of using powerful techniques such as Error Correction Coding (ECC). The project consisted of implementing ECC techniques for a Demand Access Systems (DAS) such as: Convolution codes, Reed-Solomon (R-S) codes and Concatenated codes.

First part of the project consisted of MATLAB implementation as proof of concept while the second consisted of practically relevant C++ implementation. In order to guarantee fast implementation of the second stage, parallel computing tools such multithreading where used in the C++ implementation.

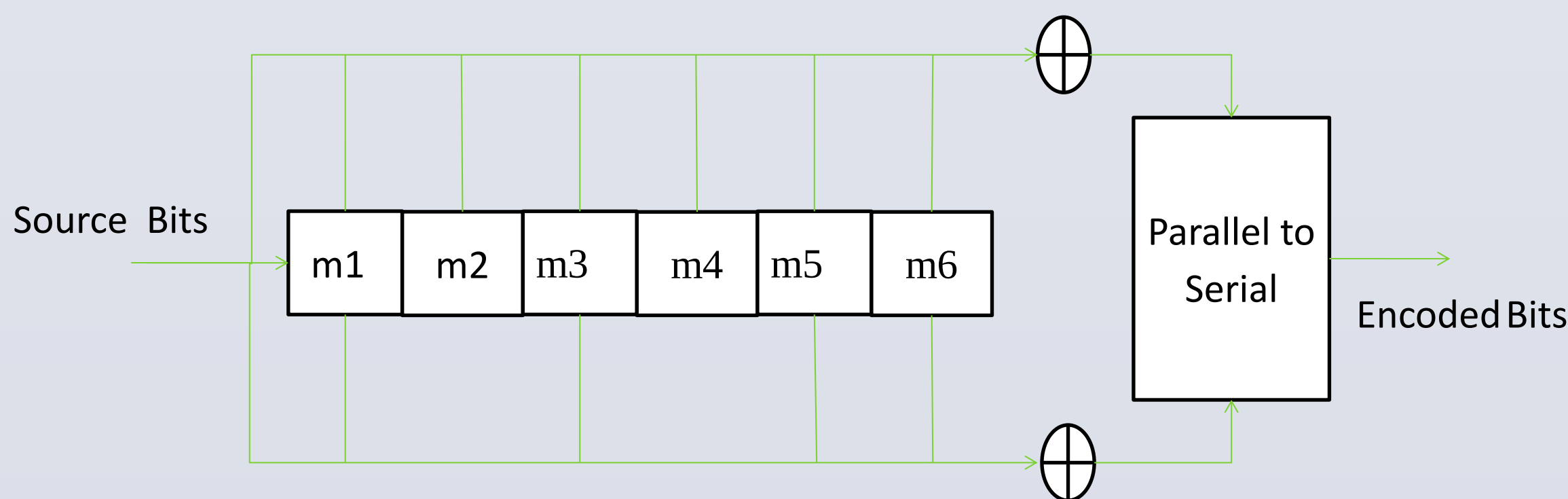
Project also provided various performance evaluations. Amongst such, is the Bit Error Rate (BER) that tells how reliable ECC is at specified Signal to Noise Ratio (SNR). Another, important matric is the Computational Complexity that describes speed and efficiency of the implementation. Project provides well documented user guided. Also, as result of our implementation we were able to compare performance of ECC used in DVB-S2 standard.

Introduction & Background

In late 1940's engineers such as Shannon and Hamming realized that in order to provide high and reliable data rates over digital channels new techniques had to be introduced and as such an area of ECC was born. ECC works by adding controllable extra redundancy to the source data (simple example is parity checks) that can compensate for the errors randomly introduced by the channel. A simple example of ECC is the famous (7,4)-Hamming code shown below where parity of each circle has even parity.

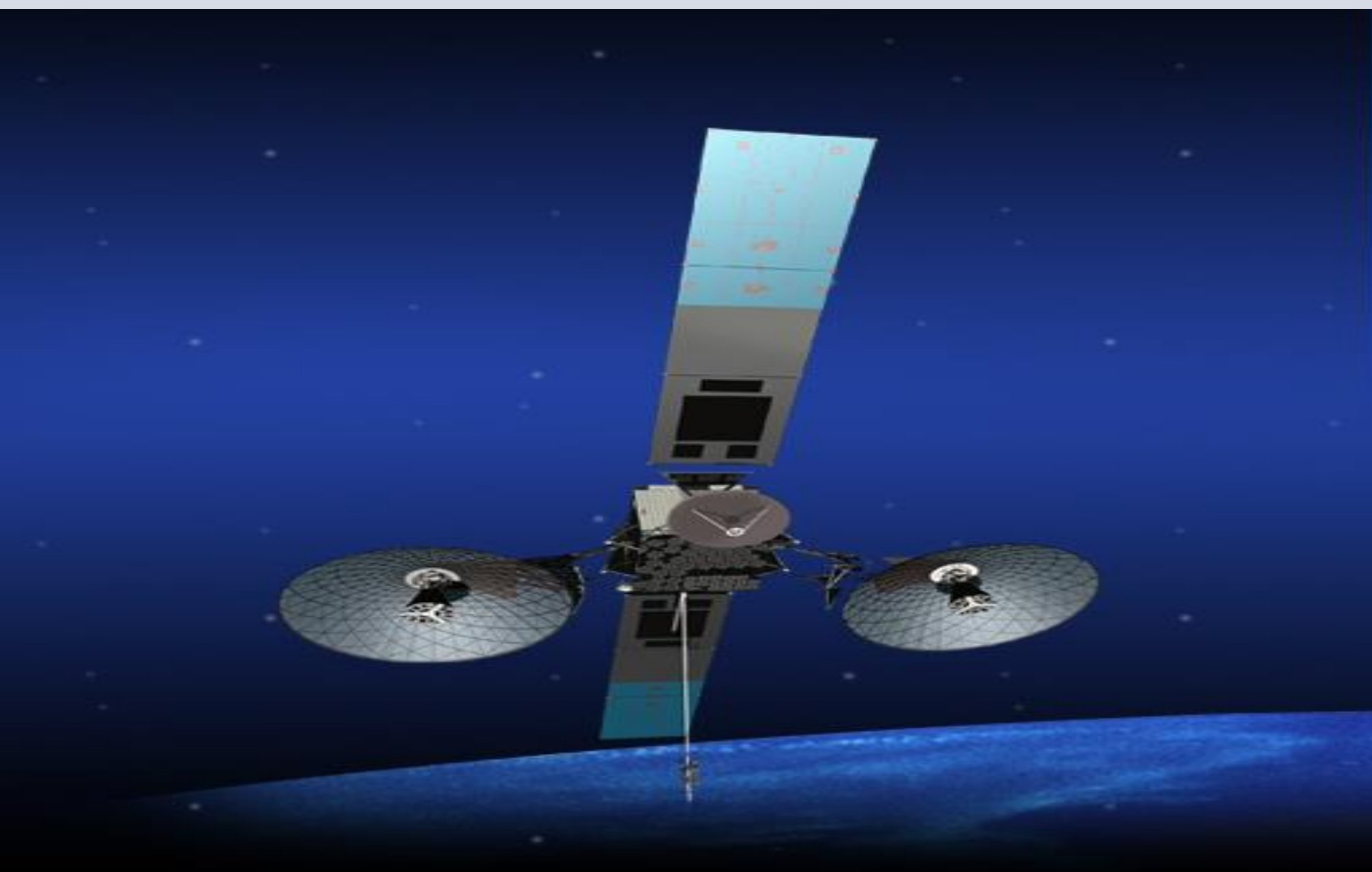


Convolution codes are a powerful error correction codes that commonly used in practice. The operation of the convolution encoder can be described by the following operation on a shift register



Once the Encoded Bits are sent through the channel, at the receiver side, a Viterbi decoding algorithm is used to recover the Source Bits.

R-S codes is another class of very powerful error correction codes. R-S codes are especially good when errors occurs in bursts. R-S codes have found wide use applications in technologies such CDs, DVDs, and Blu-Ray. R-S codes use special mathematical tools called Galois Fields which renders them very powerful.

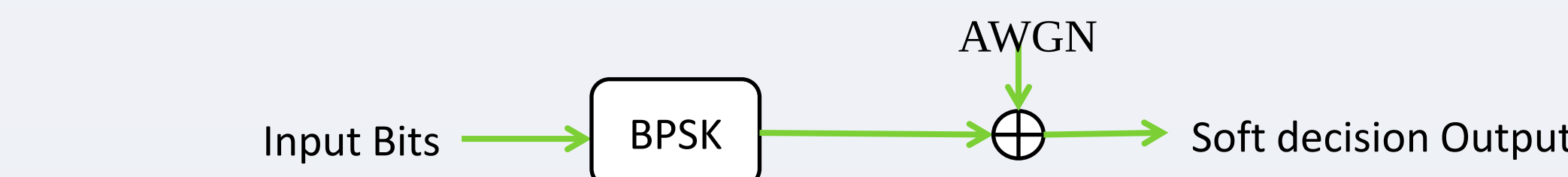


TDRSS Demand Access System (DAS) can provide variety of services to TRSS costumers. For costumers with high data requirements Convolution codes and R-S codes are a necessity. Often in order to provide even a better performance Convolution and R-S codes are combined together to form the so called concatenated code.

Implementation Details

Convolution Codes

Underplaying channel model for studying Convolution codes is the Additive White Gaussian Noise (AWGN) Channel. Input bits are mapped to BPSK symbols (0 to -1 and 1 to 1) and is send over the noisy channel.

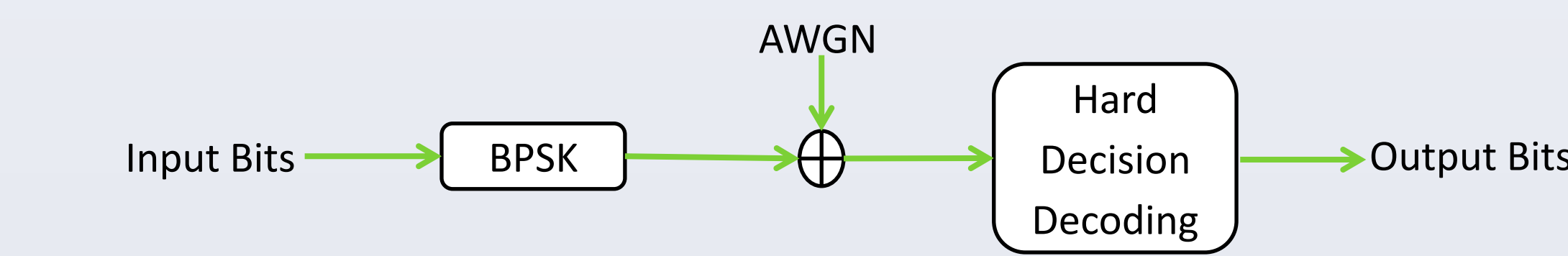


Flow diagram below shows an conceptual implementation of Convolution codes.

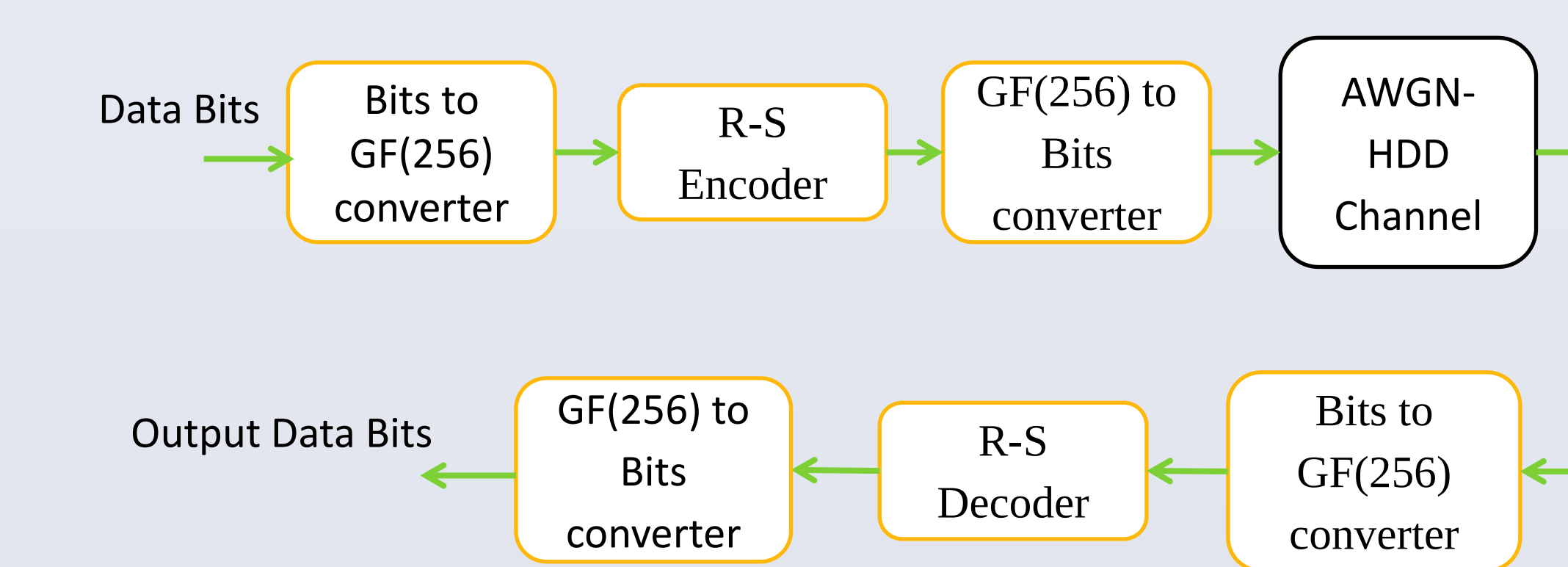


R-S Codes

The underlying channel model that is used to study performance of RS codes is AWGN channel with Hard Decision Decoding (HDD) shown below.

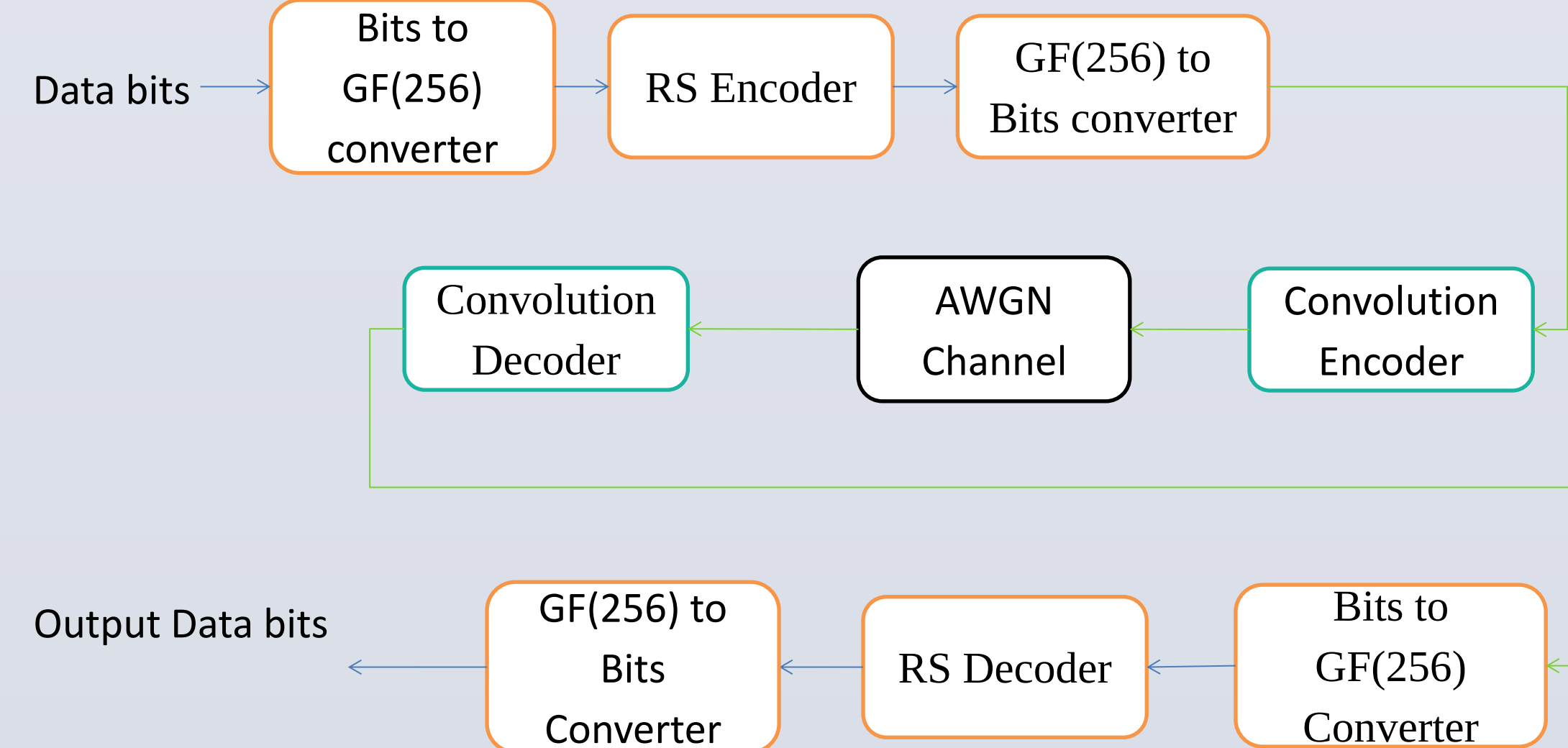


Flow diagram below shows an conceptual implementation of R-S codes.



Concatenated: Convolution + R-S codes

A high level implementation of the Concatenated code is shown below:



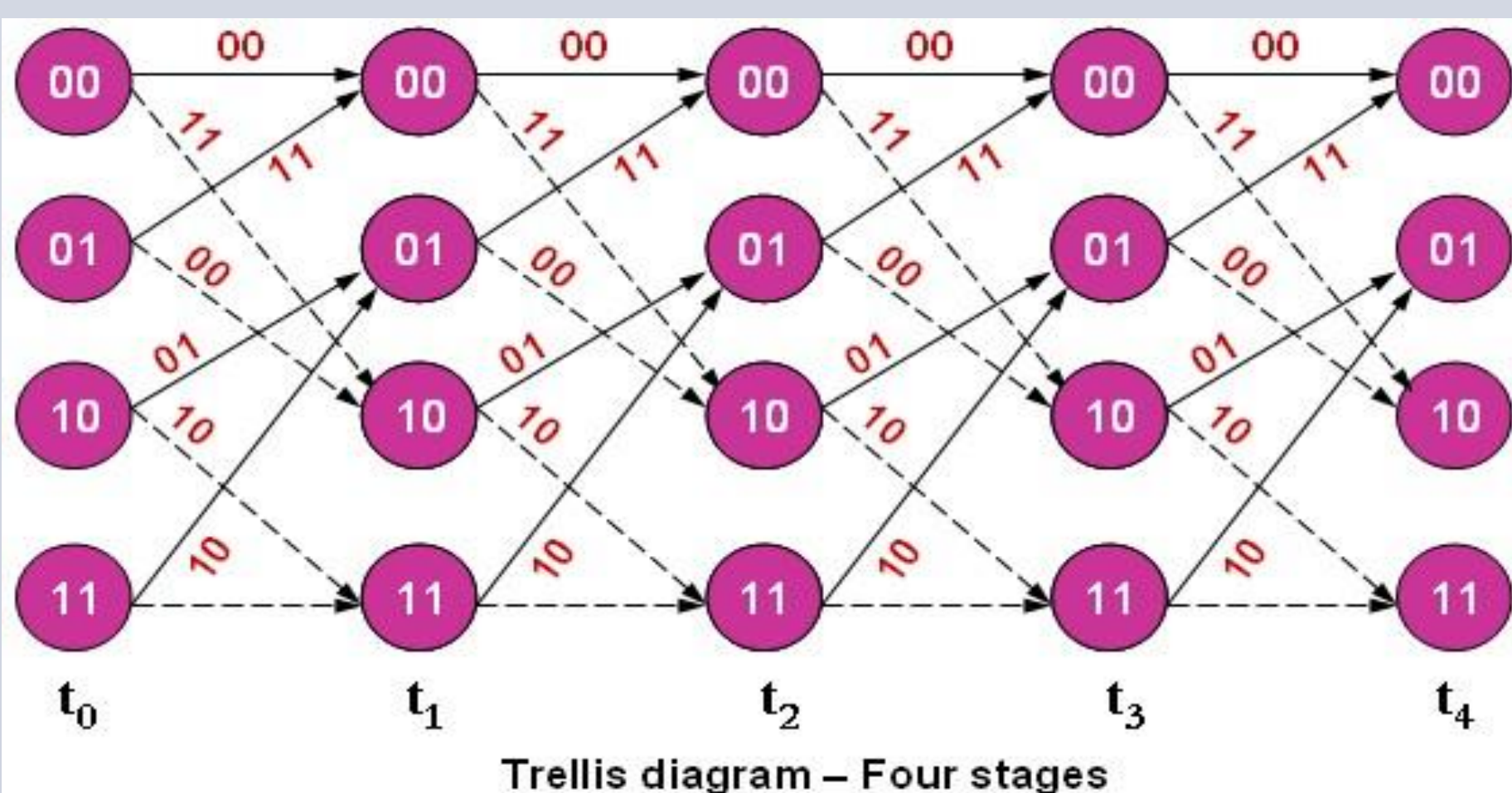
Test Implementation (MATLAB)

In order to provide a proof of concept and a working model the first stage of the project consisted of providing working solution in MATLAB. MATLAB offers an easy prototyping interface as well as useful data visualization techniques. On the other hand, MATLAB implementations are slow and are often not suitable for solutions where computational efficiency is a requirement.

Final Implementation (C++)

C++ is highly suitable for practical implementations thus the second stage of the project consisted of providing working C++ solution. In addition to provide an even better run time performance techniques such as multithreading have been used in the solution.

For example, Viterbi decoder operates by computing values of the nodes on a trellis. By running separate threads for each node we significantly reduce run time.



Another significant result of this project is a new and efficient library of function that deals with implementation Galois Fields. This library was mainly develop to handle R-S codes, however, Galois Fields are an important tool in other areas such as cryptography. Hence, the this library can be of practical interest in other fields.

Conclusions & Performance

Bit Error Rate

One of the performance metrics used the BER. Fig.1 below shows BER for uncoded data, data encoded with Convolution codes, data encoded by R-S codes and data encoded by R-S codes. Fig. 2 shows BER for as sample of codes used in DVB-S2 standard.

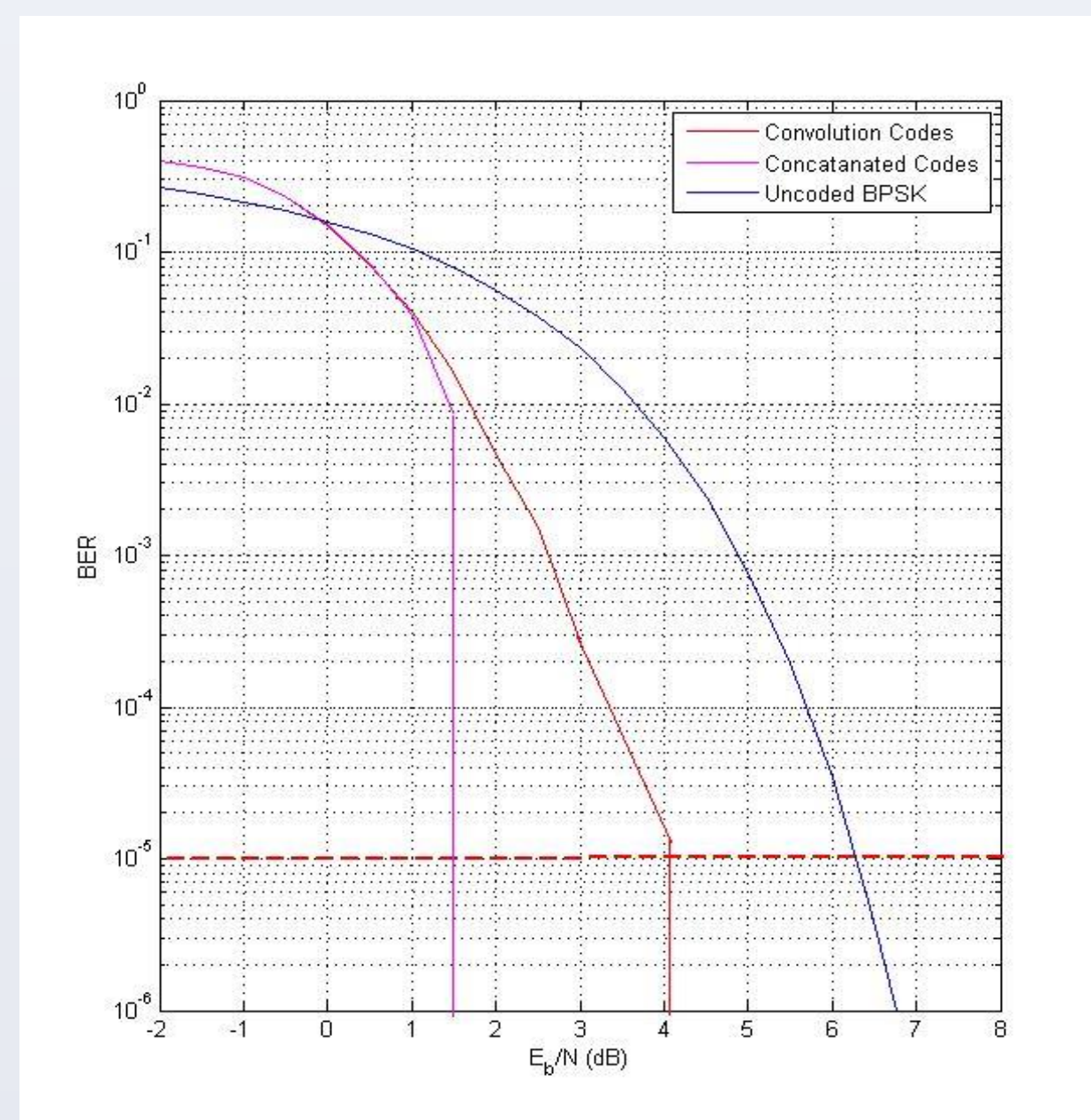


Fig.1

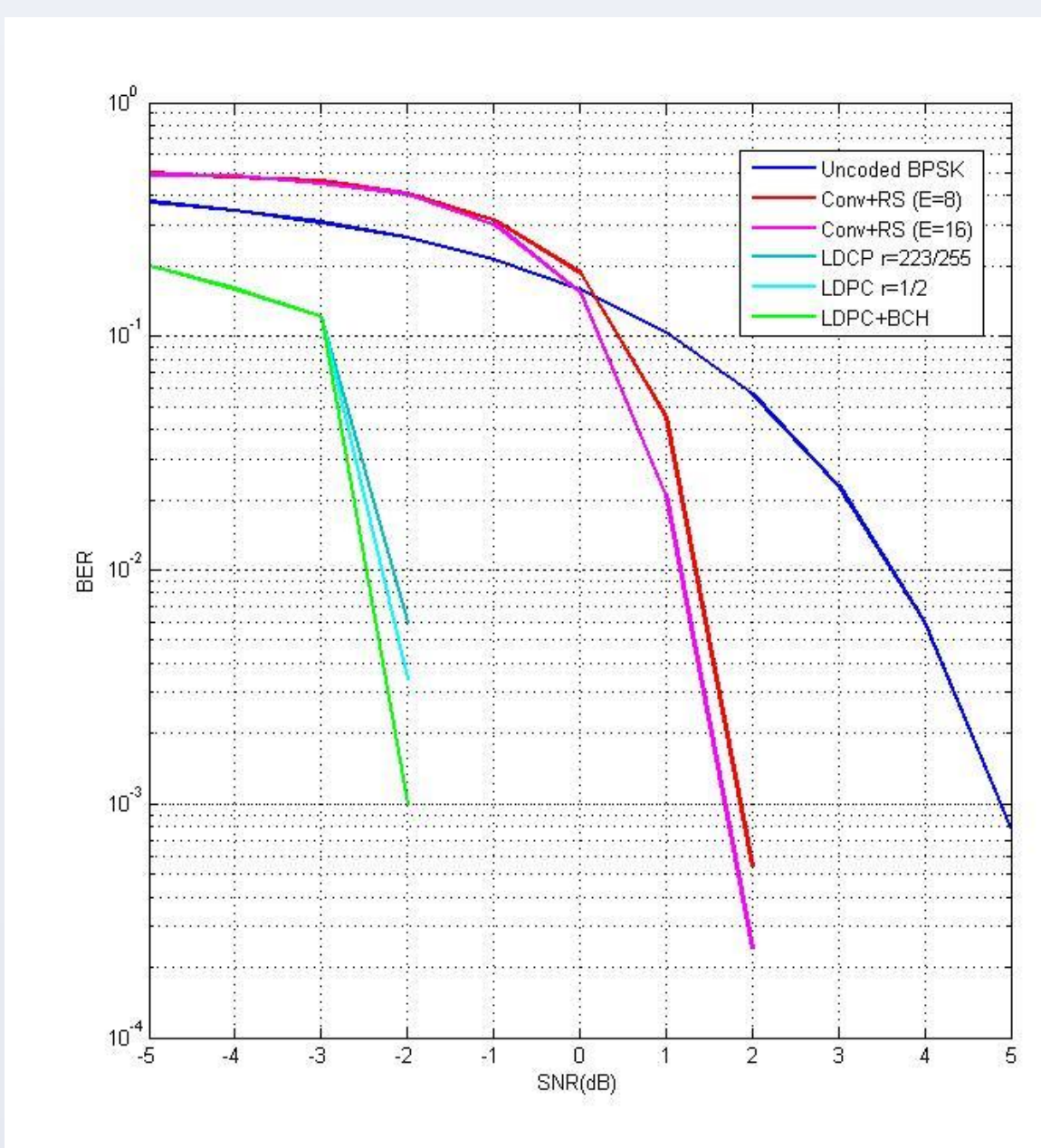


Fig. 2

Run Time

The table below compares run-times of different implementation for several block lengths

	Block Length	Time To Execute (TTE) (sec)	TTE/(base TTE)
My MATLAB Impl.	512 bits (64 bytes)	1.033376	147.6251429
	1024 bits (128 bytes)	2.142305	353.0075
	2048 bits (256 bytes)	3.889934	144.0716296
MATLAB(built in)	512 bits (64 bytes)	0.07578	10.82571429
	1024 bits (128 bytes)	0.083791	3.10337037
	2048 bits (256 bytes)	0.126071	4.669296296
C++ Impl.	512 bits (64 bytes)	0.007	1
	1024 bits (128 bytes)	0.014	1
	2048 bits (256 bytes)	0.027	1

REFERENCES

TDRSS Official Website: <http://tdrs.gsfc.nasa.gov/>

Code Specifications: http://esc.gsfc.nasa.gov/assets/files/SN_UserGuide.pdf

Error Correction Codes: Bossert, Martin. "Channel Coding for Telecommunications." New York: John Wiley & Son, 1999.

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